PCCST501 – COMPUTER NETWORKS

MODULE 1 TUTORIAL QUESTIONS 1

Q1-1. How many point-to-point WANs are needed to connect n LANs if each LAN is able to directly communicate with any other LAN?

Q1-2. When we use local telephones to talk to a friend, are we using a circuit-switched network or a packet-switched network?

Q1-3. When a resident uses a dial-up or DSL service to connect to the Internet, what is the role of the telephone company?

Q1-4. What is the first principle discussed for protocol layering that needs to be followed to make communication bidirectional?

Q1-5. Is transmission in a LAN with a common cable (Figure 1.1a) an example of broadcast (one-to-many) transmission? Explain.

Q1-6. In a LAN with a link-layer switch (Figure 1.1b), Host 1 wants to send a message to Host 3. Since communication is through the link-layer switch, does the switch need to have an address? Explain.

Q1-7. In the TCP/IP protocol suite, what are the identical objects at the sender and receiver sites when we think about the logical connection at the application layer?

Q1-8. A host communicates with another host using the TCP/IP protocol suite. What is the unit of data sent or received at each layer?

Q1-9. Which of the following data units is encapsulated in a frame?
(a) a user datagram (b) a datagram (c) a segment

Q1-10. Which of the following data units is decapsulated from a user datagram?
(a) a datagram (b) a segment (c) a message

Q1-11. Which of the following data units has an application-layer message plus the header from layer 4?
(a) a frame (b) a user datagram (c) a bit

Q1-12. List some application-layer protocols mentioned in this chapter.

Q1-13. Which layers of the TCP/IP protocol suite are involved in a link-layer switch?

Q1-14. A router connects three links (networks). How many of each of the following layers can the router be involved with?
(a) physical layer (b) data-link layer (c) network layer

Q1-15. When we say that the transport layer multiplexes and demultiplexes application-layer messages, do we mean that a transport-layer protocol can combine several messages from the application layer in one packet? Explain.

Q1-16. Can you explain why we did not mention multiplexing/demultiplexing services for the application layer?

Q1-17. If a port number is 16 bits (2 bytes), what is the minimum header size at the transport layer of the TCP/IP protocol suite?

Q1-18. What are the types of addresses (identifiers) used in each layer of the TCP/IP protocol suite?

Q1-21. Assume we want to connect two isolated hosts together to let each host communicate with the other. Do we need a link-layer switch between the two? Explain.

Q1-22. If there is a single path between the source host and the destination host, do we need a router between the two hosts?

MODULE 1 TUTORIAL PROBLEMS (P1-1 TO P1-16)

P1-1. Assume that the number of hosts connected to the Internet in 2010 is five hundred million. If the number of hosts increases by 20% per year, what is the number of hosts in 2020?

P1-2. Assume that a system uses five protocol layers. If the application program creates a message of 100 bytes and each layer (including the fifth and the first) adds a header of 10 bytes to the data unit, what is the efficiency (the ratio of application-layer bytes to the number of bytes transmitted) of the system?

P1-3. Answer the following questions about Figure 1.10 when communication is from Maria to Ann:
(a) What service is provided by layer 1 to layer 2 at Maria's site?
(b) What service is provided by layer 1 to layer 2 at Ann's site?

P1-4. Answer the following questions about Figure 1.10 when communication is from Maria to Ann:
(a) What service is provided by layer 2 to layer 3 at Maria's site?
(b) What service is provided by layer 2 to layer 3 at Ann's site?

P1-5. Match the following to one or more layers of the TCP/IP protocol suite:
(a) creating user datagrams
(b) responsibility for handling frames between adjacent nodes
(c) transforming bits to electromagnetic signals

P1-6. In Figure 1.18, when the IP protocol decapsulates the transport-layer packet, how does it know to which upper-layer protocol (UDP or TCP) the packet should be delivered?

P1-7. Assume a private internet uses three different protocols at the data-link layer (L1, L2, and L3). Redraw Figure 1.18 with this assumption. Can we say that, in the data-link layer, we have demultiplexing at the source node and multiplexing at the destination node?

P1-8. Assume that a private internet requires messages at the application layer to be encrypted and decrypted for security. If we need to add information about the encryption/decryption process, such as the algorithms used, does it mean that we are adding one layer to the TCP/IP protocol suite? Redraw the TCP/IP layers (Figure 1.12b) if you think so.

P1-9. Assume that we have created a packet-switched internet. Using the TCP/IP protocol suite, we need to transfer a huge file. What is the advantage and disadvantage of sending large packets?

P1-10. Match the following to one or more layers of the TCP/IP protocol suite:
(a) route determination
(b) connection to transmission media
(c) providing services for the end user

P1-11. The presentation of data is becoming increasingly important in today's Internet. Some people argue that the TCP/IP protocol suite needs a new layer to handle data presentation. If this layer is added, where should it be positioned in the suite? Redraw Figure 1.12 to include this layer.

P1-12. In an internet, we change the LAN technology to a new one. Which layers in the TCP/IP protocol suite need to be changed?

P1-13. Assume that an application-layer protocol is written to use the services of UDP. Can the application-layer protocol use the services of TCP without change?

P1-14. Using the internet in Figure 1.4, show the layers of the TCP/IP protocol suite and the flow of data when two hosts, one on the west coast and the other on the east coast, exchange messages.

P1-15. Protocol layering can be found in many aspects of life, such as air travel. Imagine you make a round trip to spend a vacation at a resort. Show the protocol layering for the round trip using layers such as baggage checking/claiming, boarding/unboarding, and takeoff/landing.

P1-16. In Figure 1.4, there is only a single path from a host at the west coast office to a host at the east coast office. Why do we need two routers in this internetwork?